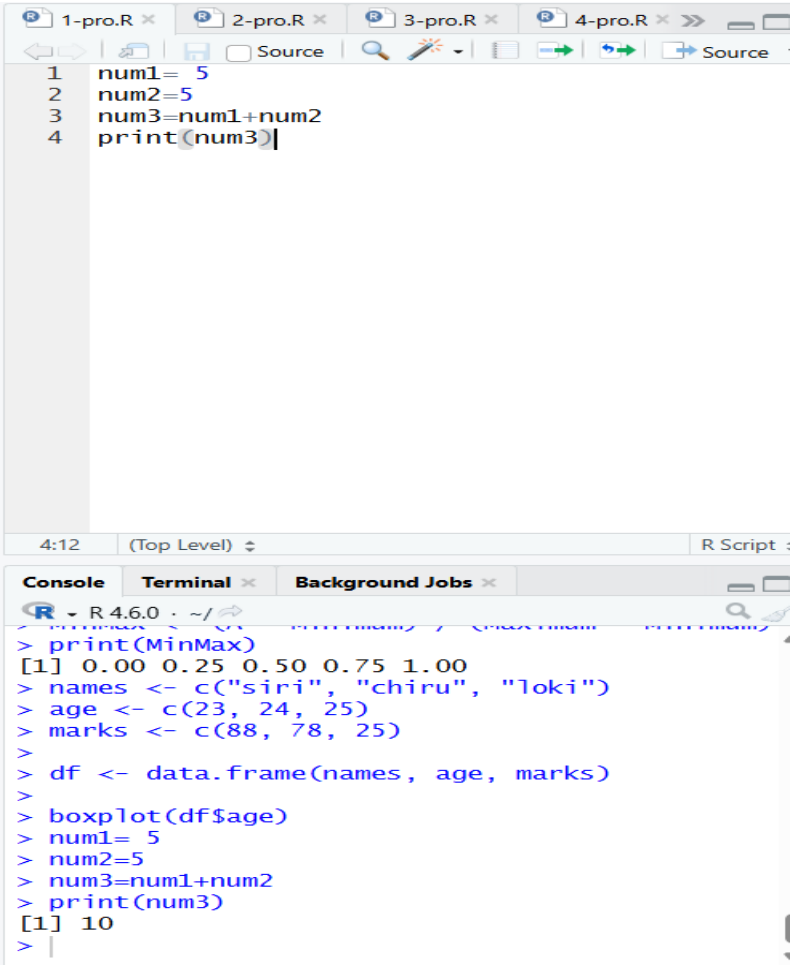
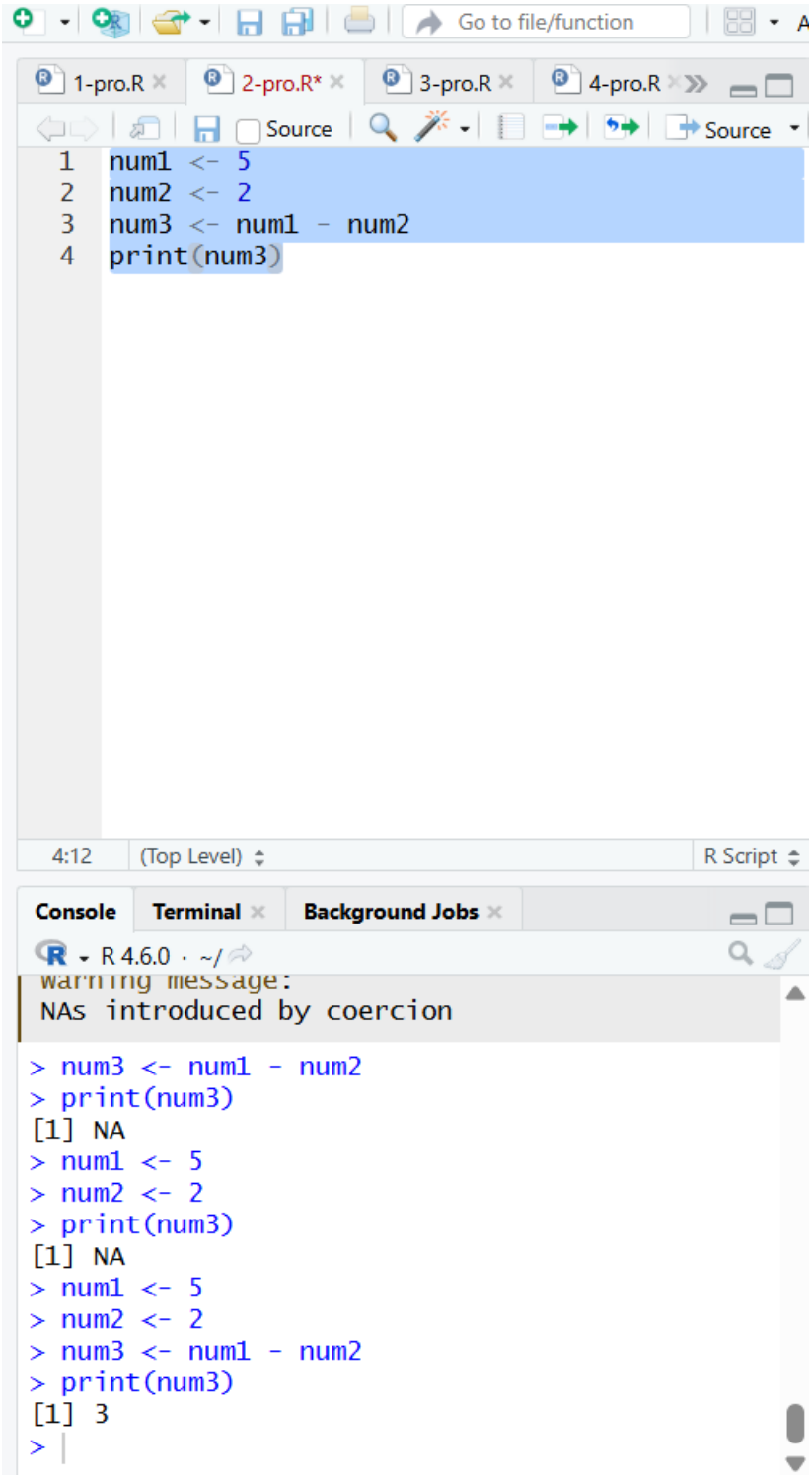
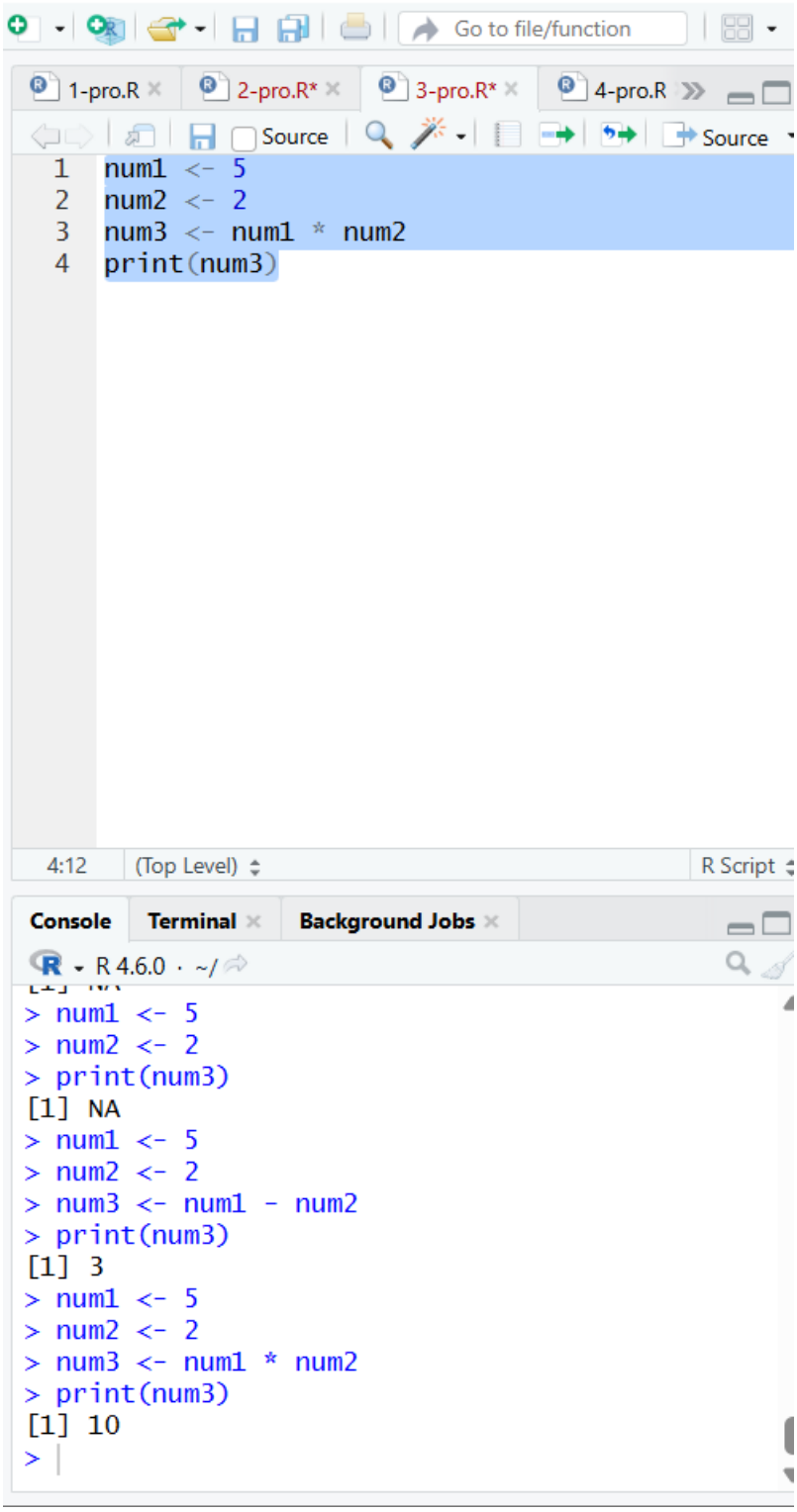
EXP-1


1-pro.R
2-pro.R
3-pro.R
4-pro.R
Source
Source
1 num1= 5
2 num2=5
3 num3=num1+num2
4 print(num3)
4:12
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> print(MinMax)
[1] 0.00 0.25 0.50 0.75 1.00
> names <- c("siri", "chiru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> boxplot(df$age)
> num1= 5
> num2=5
> num3=num1+num2
> print(num3)
[1] 10
>

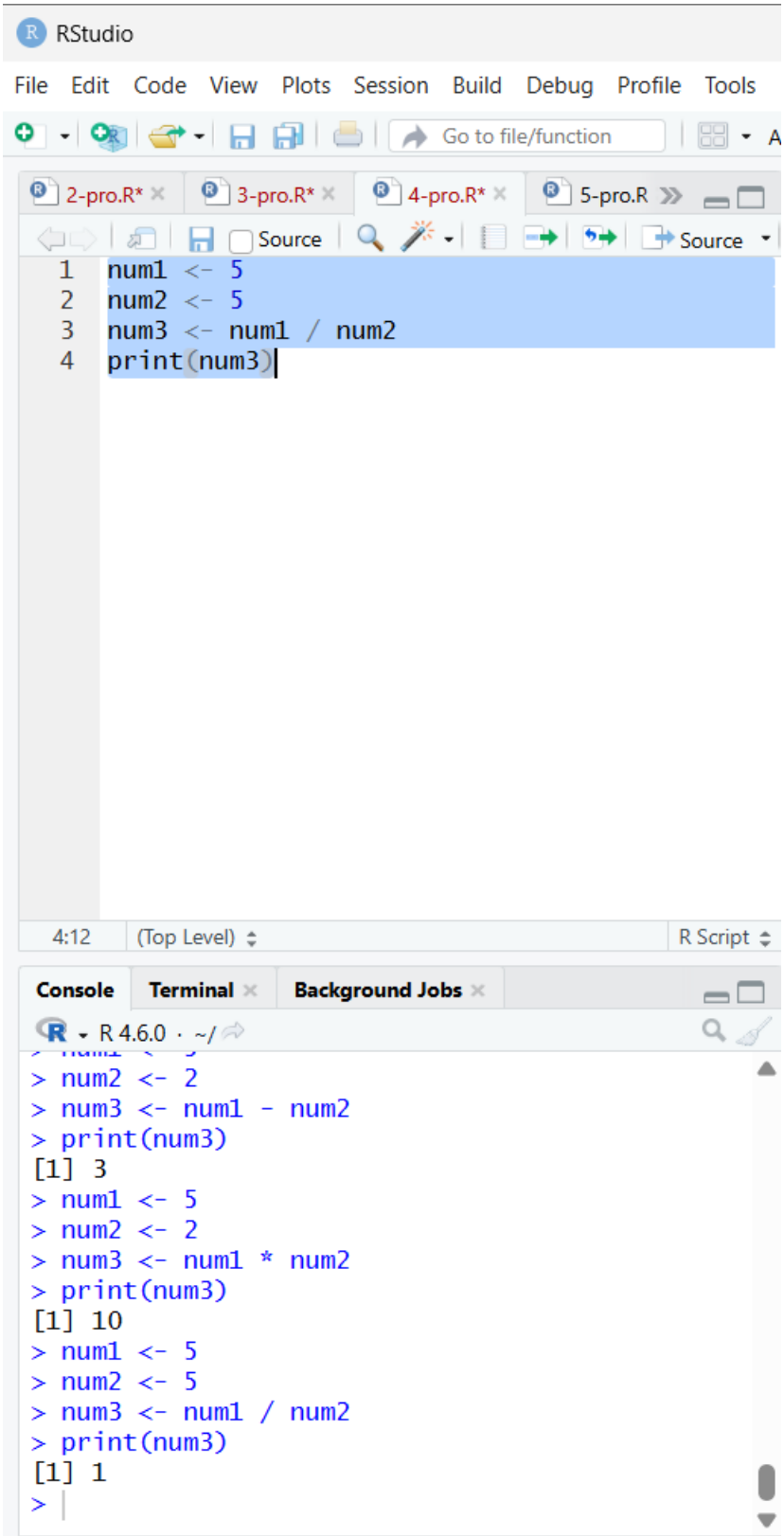
EXP-2


Go to file/function
1-pro.R
2-pro.R*
3-pro.R
4-pro.R
Source
Source
1 num1 <- 5
2 num2 <- 2
3 num3 <- num1 - num2
4 print(num3)
4:12
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
Warning message:
NAs introduced by coercion
> num3 <- num1 - num2
> print(num3)
[1] NA
> num1 <- 5
> num2 <- 2
> print(num3)
[1] NA
> num1 <- 5
> num2 <- 2
> num3 <- num1 - num2
> print(num3)
[1] 3
>

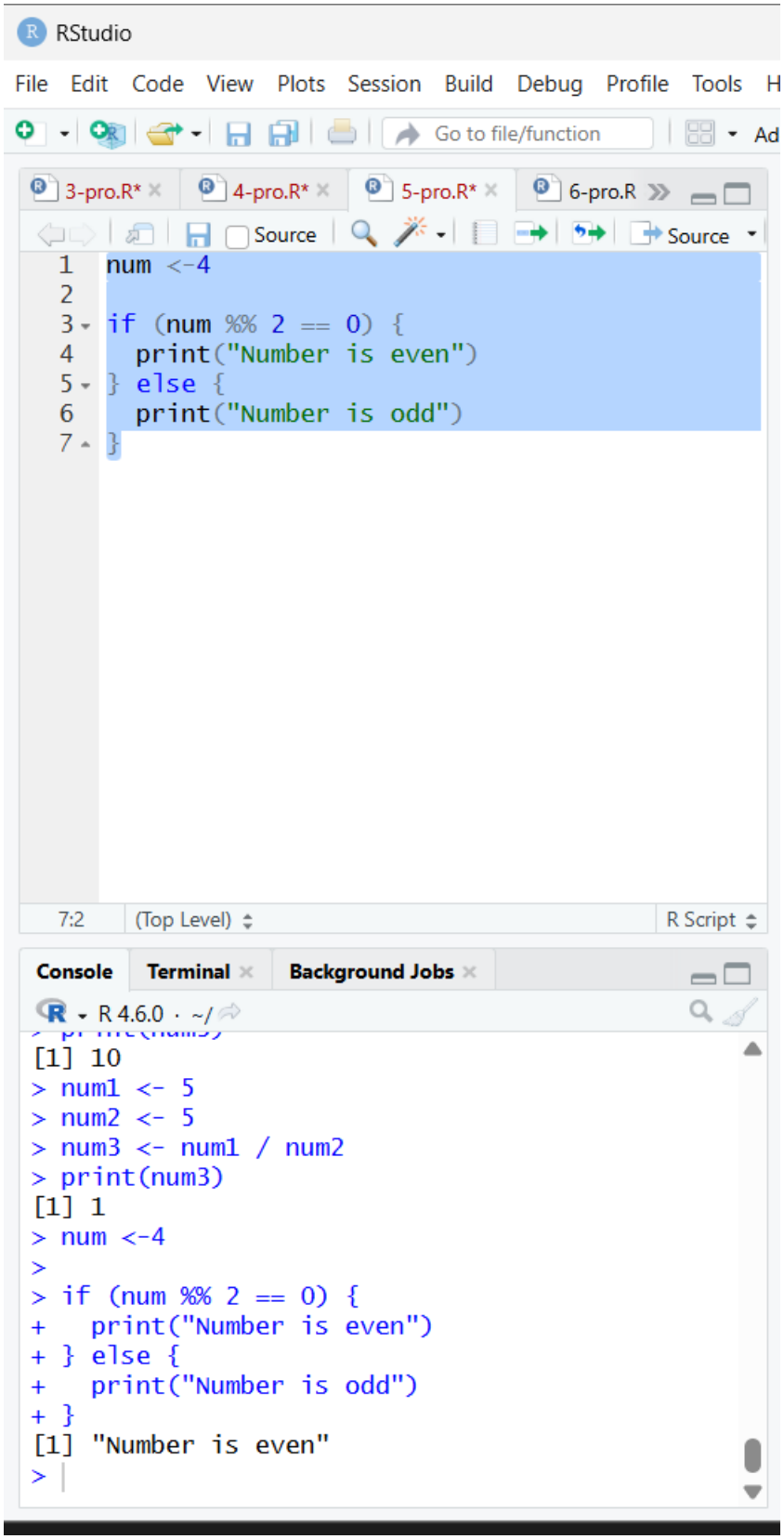
EXP-3


Go to file/function
1-pro.R
2-pro.R*
3-pro.R*
4-pro.R
Source
Source
1 num1 <- 5
2 num2 <- 2
3 num3 <- num1 * num2
4 print(num3)
4:12
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> num1 <- 5
> num2 <- 2
> print(num3)
[1] NA
> num1 <- 5
> num2 <- 2
> num3 <- num1 - num2
> print(num3)
[1] 3
> num1 <- 5
> num2 <- 2
> num3 <- num1 * num2
> print(num3)
[1] 10
>

EXP-4


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
2-pro.R*
3-pro.R*
4-pro.R*
5-pro.R
Source
Source
1 num1 <- 5
2 num2 <- 5
3 num3 <- num1 / num2
4 print(num3)
4:12
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> num2 <- 2
> num3 <- num1 - num2
> print(num3)
[1] 3
> num1 <- 5
> num2 <- 2
> num3 <- num1 * num2
> print(num3)
[1] 10
> num1 <- 5
> num2 <- 5
> num3 <- num1 / num2
> print(num3)
[1] 1
>

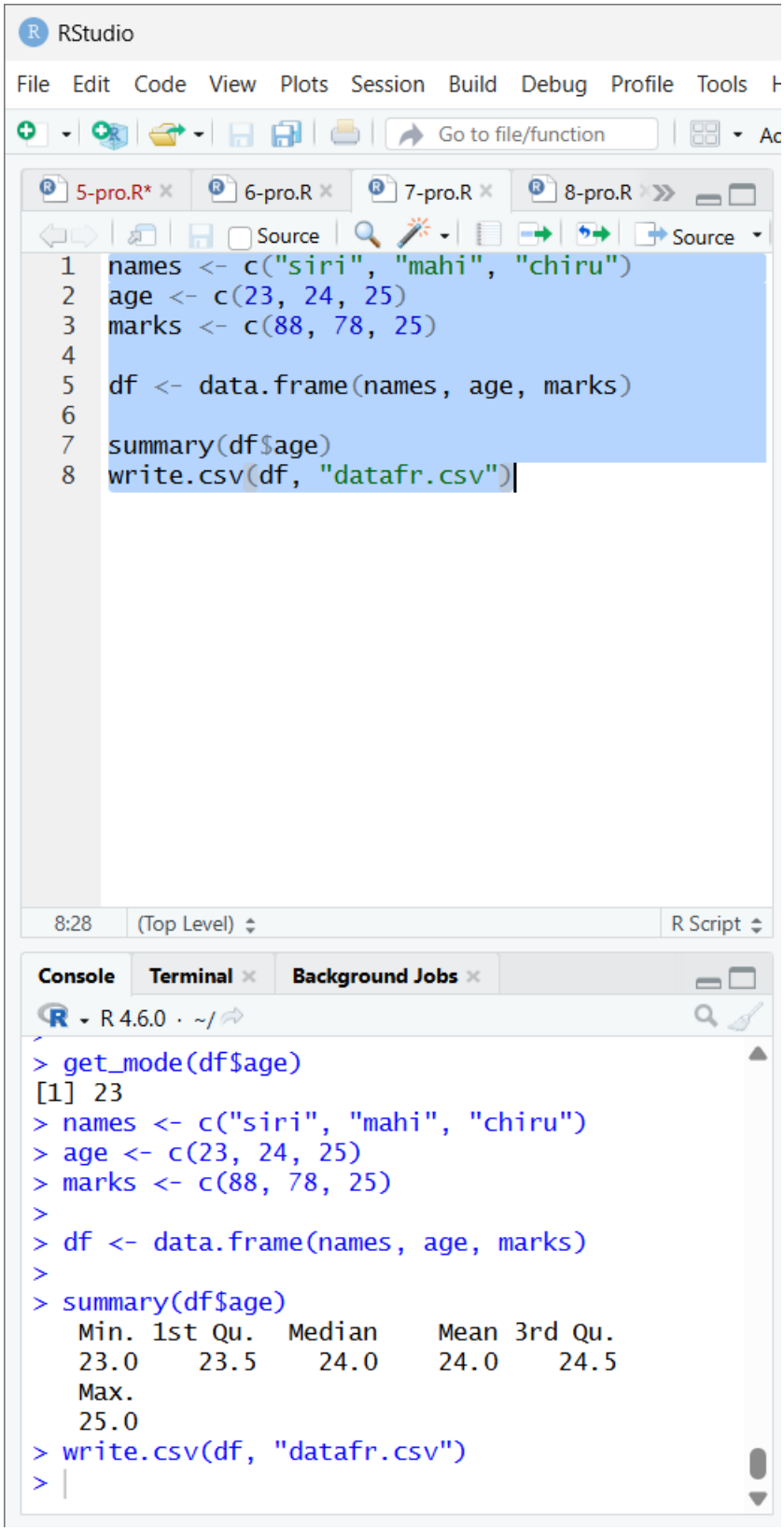
EXP-5


RStudio
File Edit Code View Plots Session Build Debug Profile Tools H
Go to file/function
3-pro.R*
4-pro.R*
5-pro.R*
6-pro.R
Source
Source
num <-4
if (num %% 2 == 0) {
print("Number is even")
} else {
print("Number is odd")
}
7:2
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
[1] 10
> num1 <- 5
> num2 <- 5
> num3 <- num1 / num2
> print(num3)
[1] 1
> num <-4
>
> if (num %% 2 == 0) {
+ print("Number is even")
+ } else {
+ print("Number is odd")
+ }
[1] "Number is even"
>

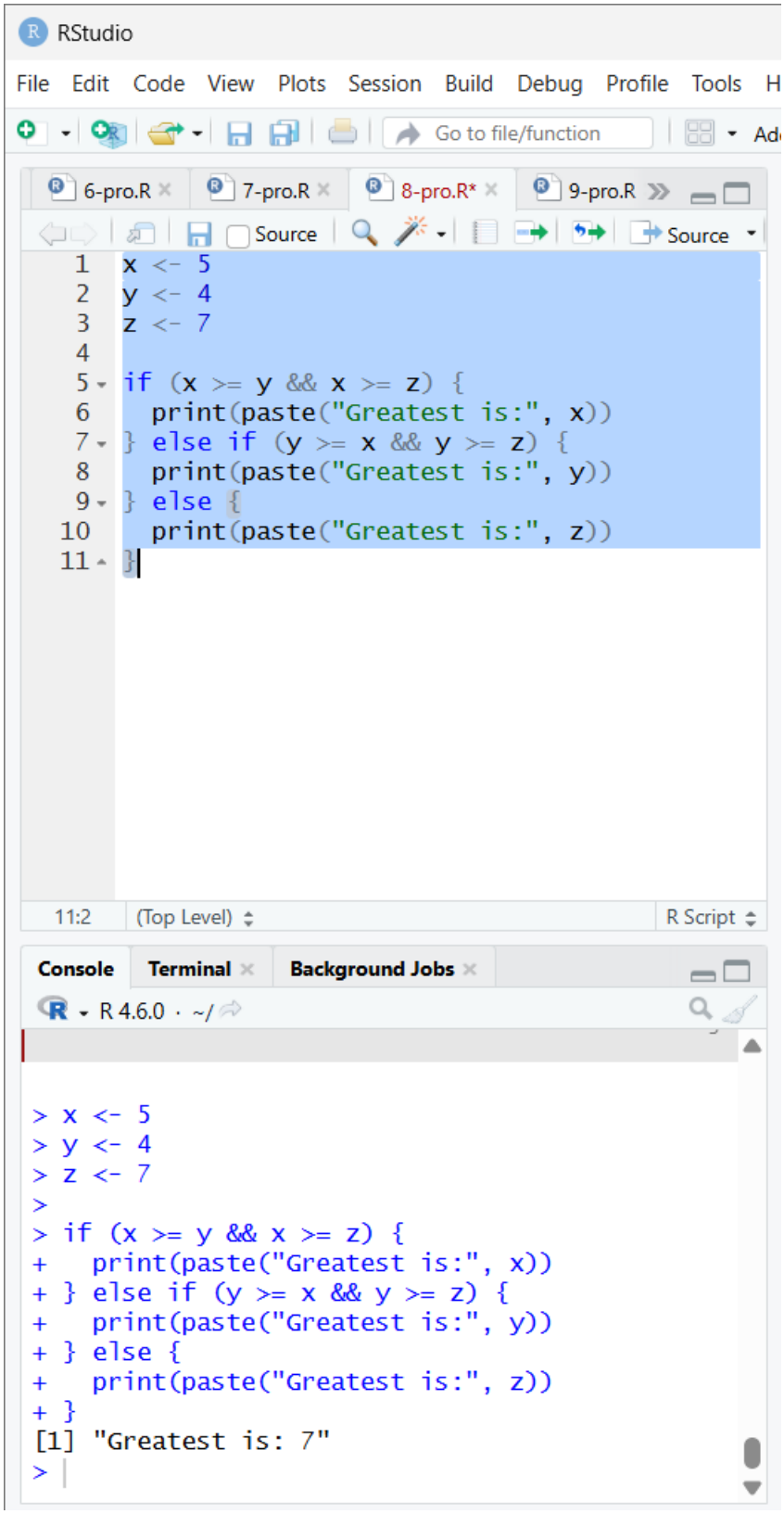
EXP-6


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
4-pro.R*
5-pro.R*
6-pro.R
7-pro.R
Source
Source
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 mean(df$age)
8 median(df$age)
9
10 get_mode <- function(x) {
11 values <- unique(x)
12 values[which.max(tabulate(match(x, values
13 }
14
15 get_mode(df$age)
15:17
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> mean(df$age)
[1] 24
> median(df$age)
[1] 24
> get_mode <- function(x) {
+ values <- unique(x)
+ values[which.max(tabulate(match(x, value
s)))]
+ }
> get_mode(df$age)
[1] 23

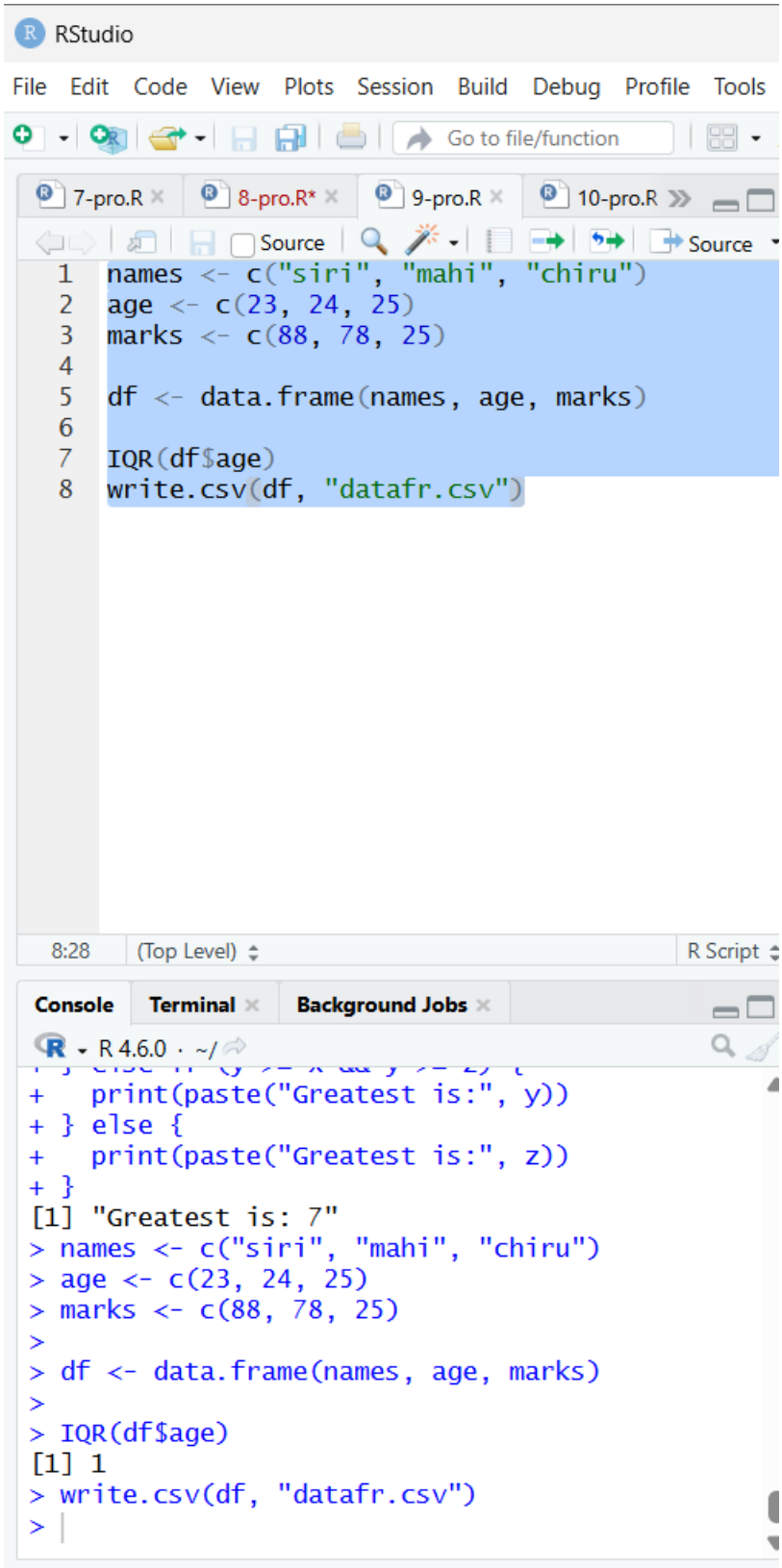
EXP-7


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
5-pro.R*
6-pro.R
7-pro.R
8-pro.R
Source
Source
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 summary(df$age)
8 write.csv(df, "datafr.csv")
8:28
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> get_mode(df$age)
[1] 23
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> summary(df$age)
Min. 1st Qu. Median Mean 3rd Qu.
23.0 23.5 24.0 24.0 24.5
Max.
25.0
> write.csv(df, "datafr.csv")
>

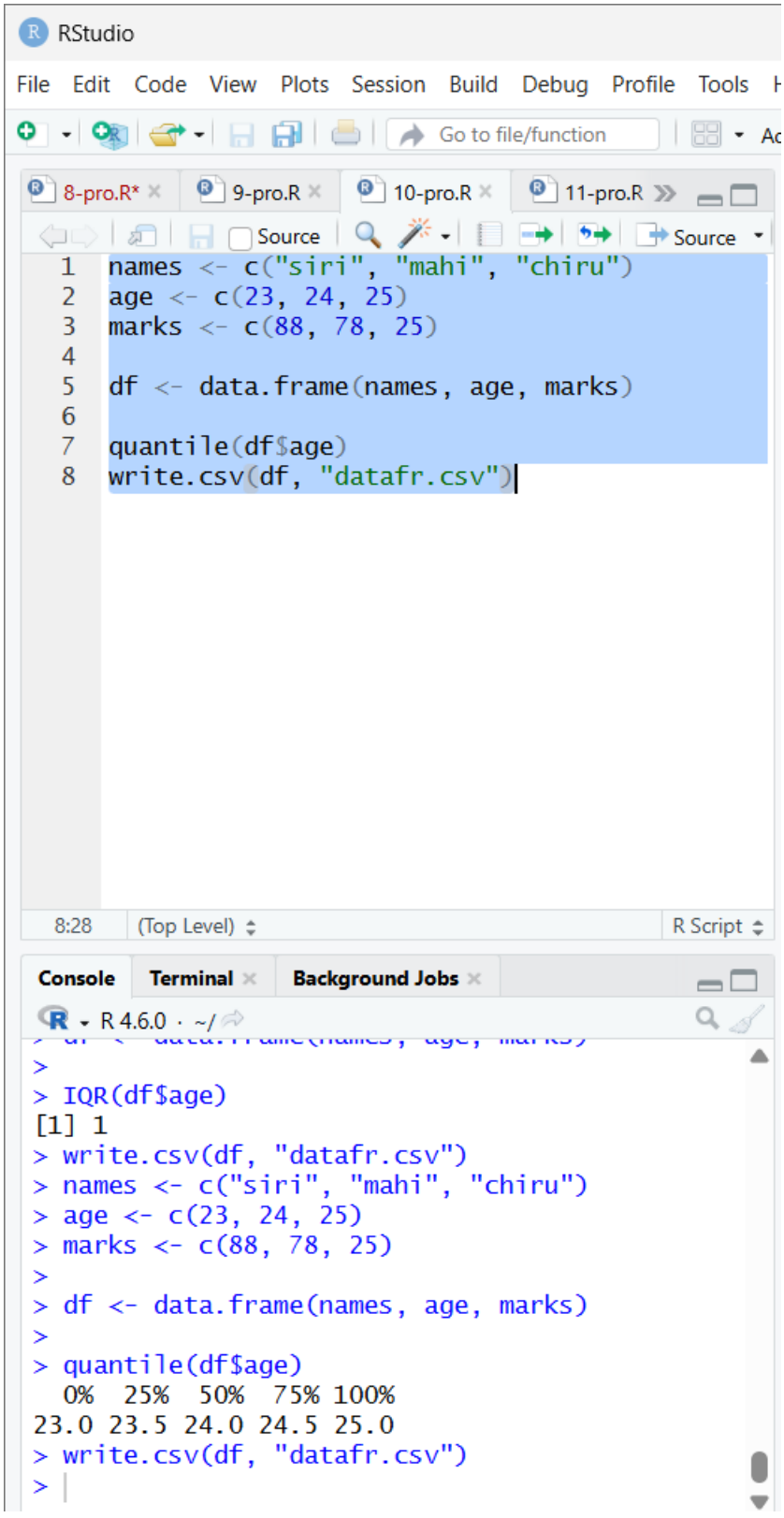
EXP-8


RStudio
File Edit Code View Plots Session Build Debug Profile Tools H
Go to file/function
6-pro.R
7-pro.R
8-pro.R*
9-pro.R
Source
Source
x <- 5
y <- 4
z <- 7
if (x >= y && x >= z) {
print(paste("Greatest is:", x))
} else if (y >= x && y >= z) {
print(paste("Greatest is:", y))
} else {
print(paste("Greatest is:", z))
}
11:2
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> x <- 5
> y <- 4
> z <- 7
>
> if (x >= y && x >= z) {
+ print(paste("Greatest is:", x))
+ } else if (y >= x && y >= z) {
+ print(paste("Greatest is:", y))
+ } else {
+ print(paste("Greatest is:", z))
+ }
[1] "Greatest is: 7"
>

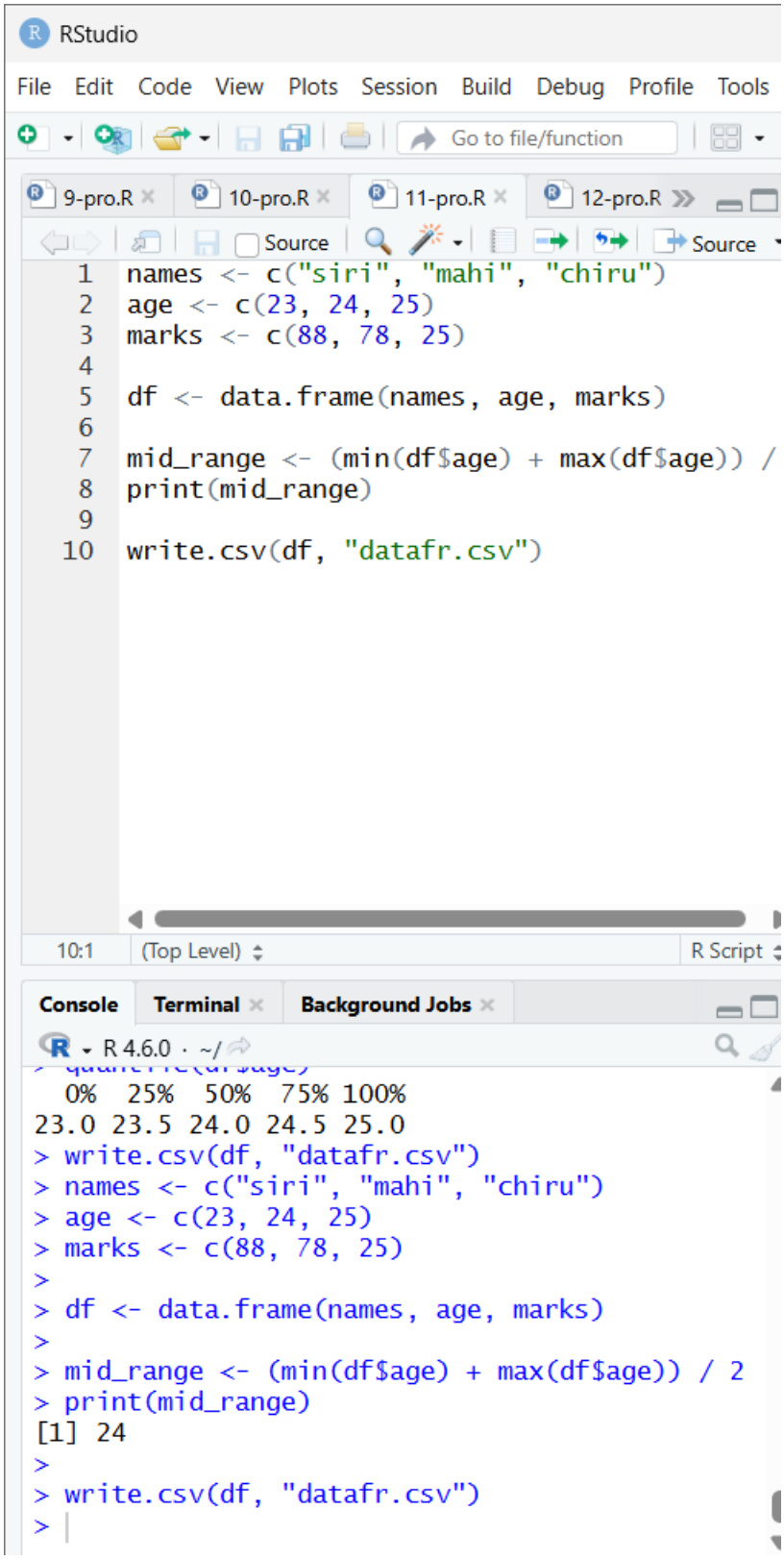
EXP-9


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
7-pro.R
8-pro.R*
9-pro.R
10-pro.R
Source
Source
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 IQR(df$age)
8 write.csv(df, "datafr.csv")
8:28
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
+ print(paste("Greatest is:", y))
+ } else {
+ print(paste("Greatest is:", z))
+ }
[1] "Greatest is: 7"
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> IQR(df$age)
[1] 1
> write.csv(df, "datafr.csv")
>

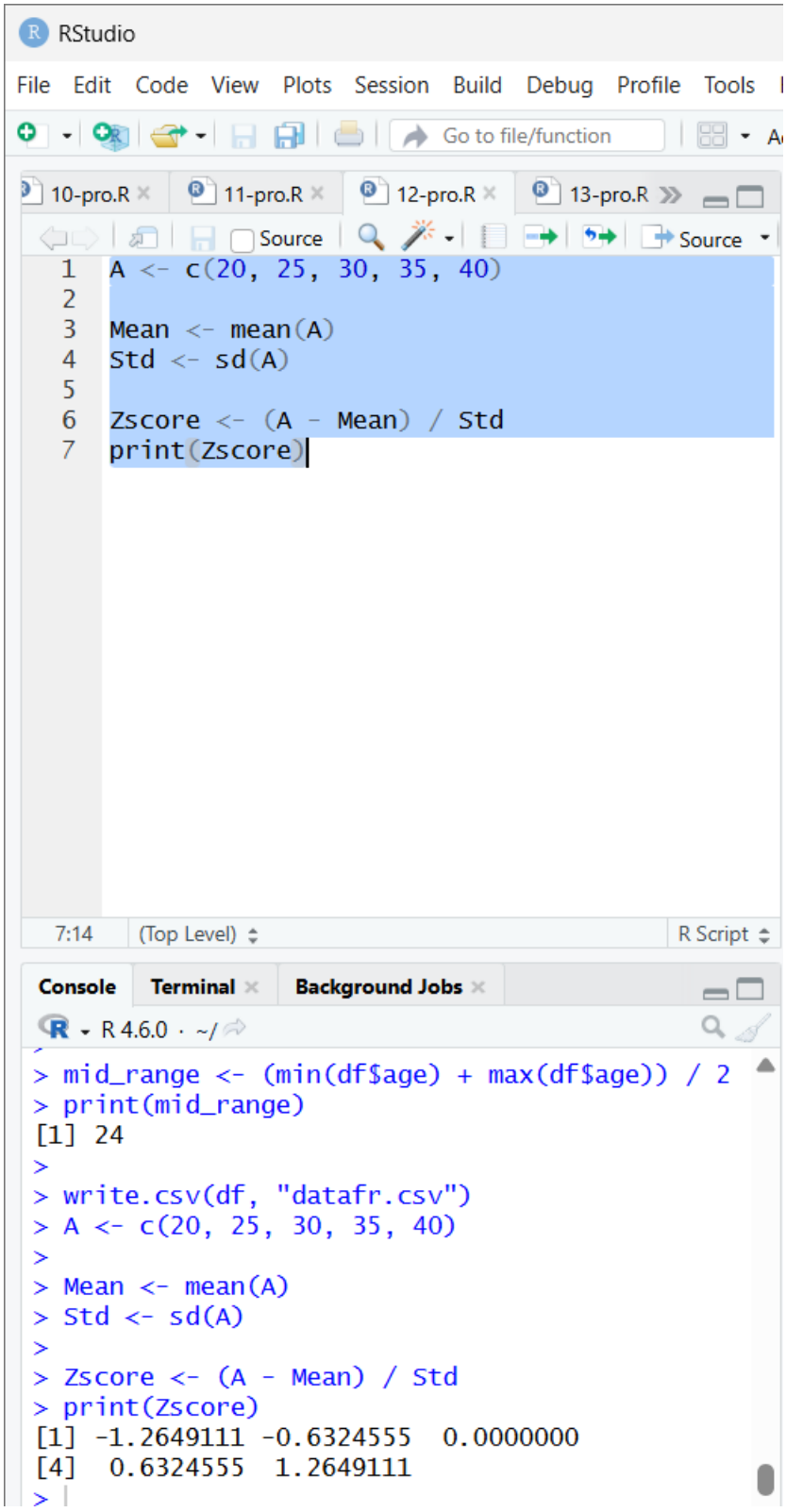
EXP-10


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
8-pro.R*
9-pro.R
10-pro.R
11-pro.R
Source
Source
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 quantile(df$age)
8 write.csv(df, "datafr.csv")
8:28
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> IQR(df$age)
[1] 1
> write.csv(df, "datafr.csv")
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> quantile(df$age)
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv")

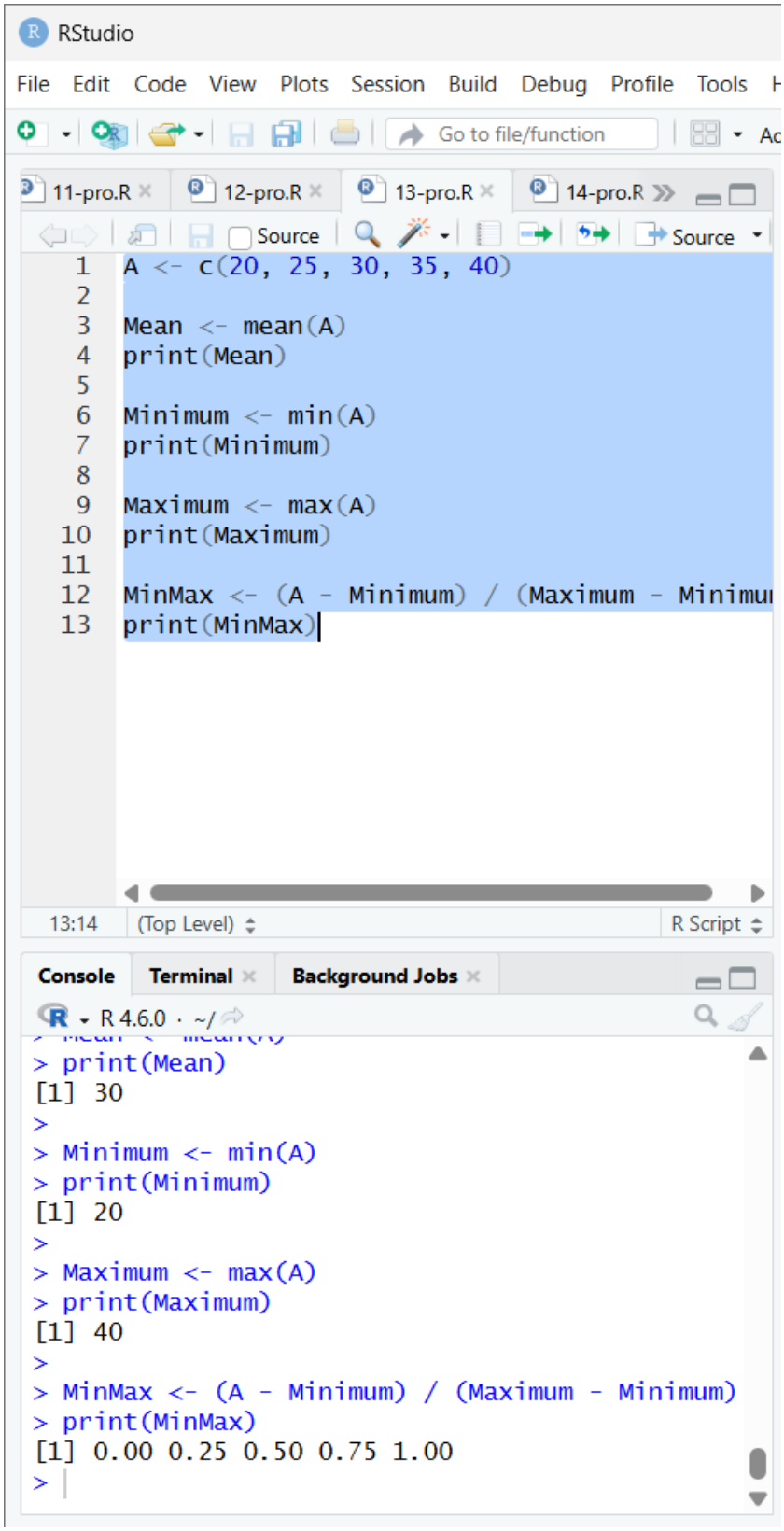
EXP-11


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
9-pro.R
10-pro.R
11-pro.R
12-pro.R
Source
Source
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 mid_range <- (min(df$age) + max(df$age)) /
8 print(mid_range)
9
10 write.csv(df, "datafr.csv")
10:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
0% 25% 50% 75% 100%
23.0 23.5 24.0 24.5 25.0
> write.csv(df, "datafr.csv")
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
>
> write.csv(df, "datafr.csv")
>

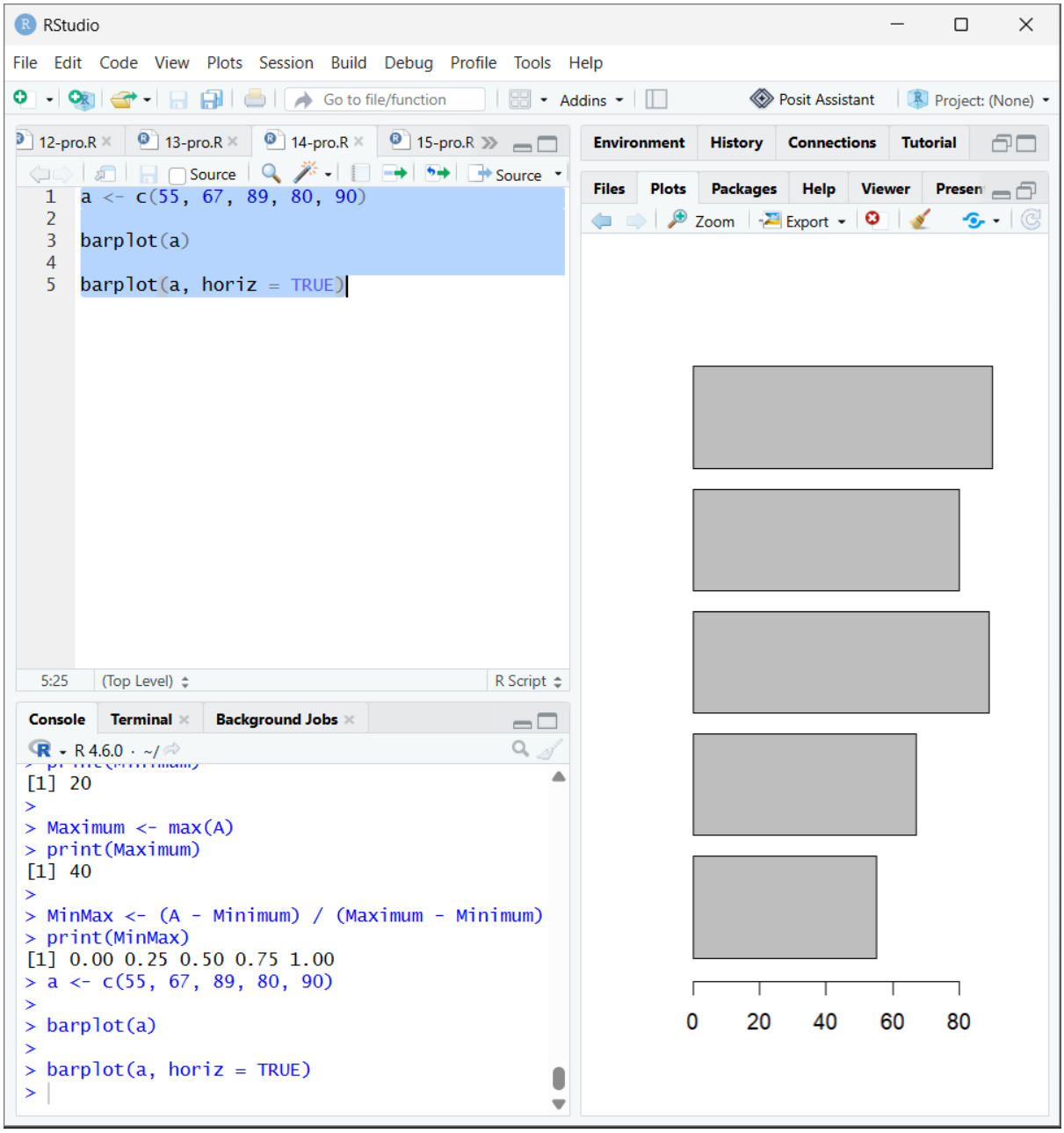
EXP-12


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
10-pro.R
11-pro.R
12-pro.R
13-pro.R
Source
Source
1 A <- c(20, 25, 30, 35, 40)
2
3 Mean <- mean(A)
4 Std <- sd(A)
5
6 Zscore <- (A - Mean) / Std
7 print(Zscore)
7:14
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> mid_range <- (min(df$age) + max(df$age)) / 2
> print(mid_range)
[1] 24
>
> write.csv(df, "datafr.csv")
> A <- c(20, 25, 30, 35, 40)
>
> Mean <- mean(A)
> Std <- sd(A)
>
> Zscore <- (A - Mean) / Std
> print(Zscore)
[1] -1.2649111 -0.6324555 0.0000000
[4] 0.6324555 1.2649111
>

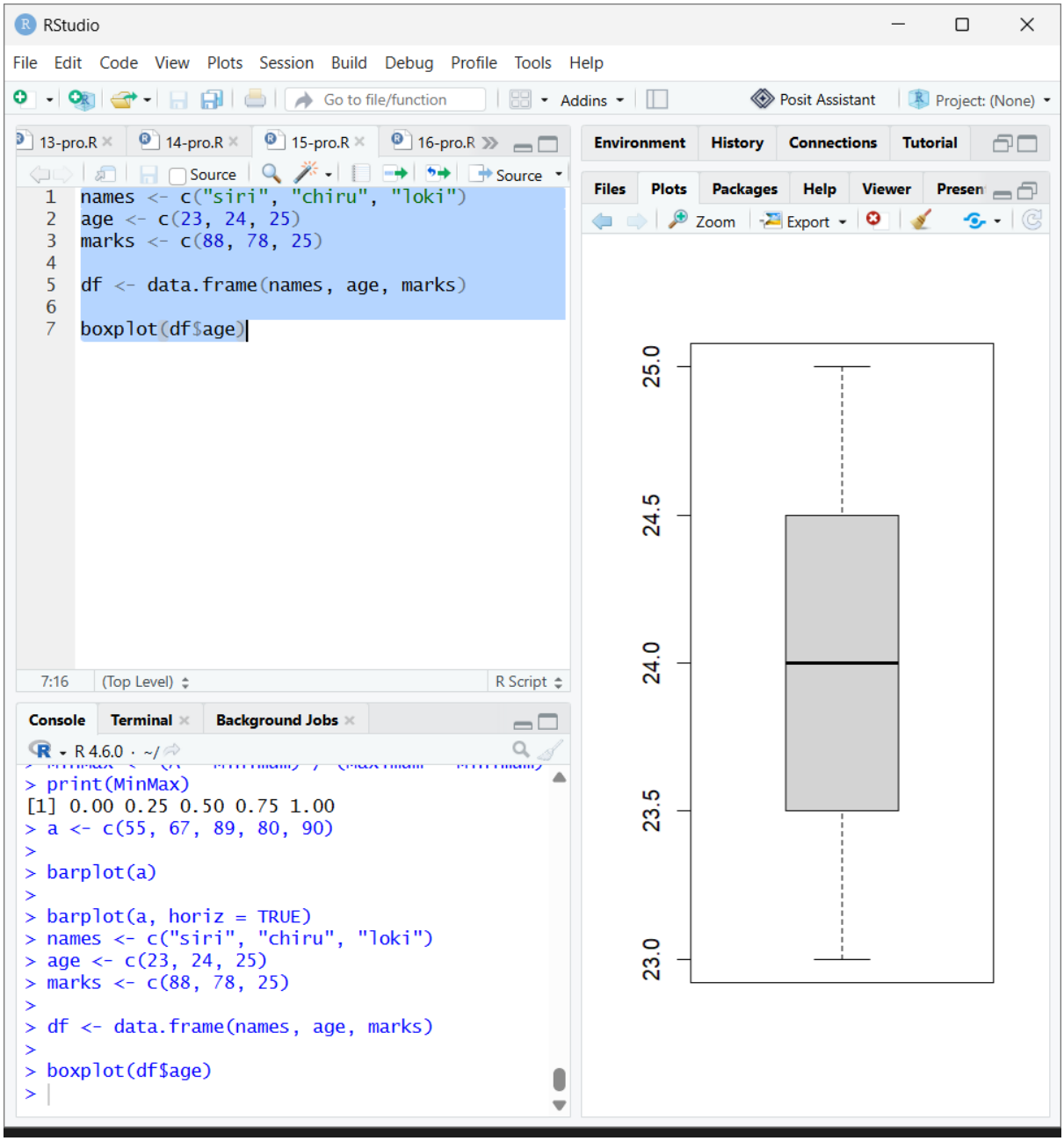
EXP-13


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
11-pro.R
12-pro.R
13-pro.R
14-pro.R
Source
Source
1 A <- c(20, 25, 30, 35, 40)
2
3 Mean <- mean(A)
4 print(Mean)
5
6 Minimum <- min(A)
7 print(Minimum)
8
9 Maximum <- max(A)
10 print(Maximum)
11
12 MinMax <- (A - Minimum) / (Maximum - Minimu
13 print(MinMax)
13:14
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> print(Mean)
[1] 30
>
> Minimum <- min(A)
> print(Minimum)
[1] 20
>
> Maximum <- max(A)
> print(Maximum)
[1] 40
>
> MinMax <- (A - Minimum) / (Maximum - Minimum)
> print(MinMax)
[1] 0.00 0.25 0.50 0.75 1.00
>

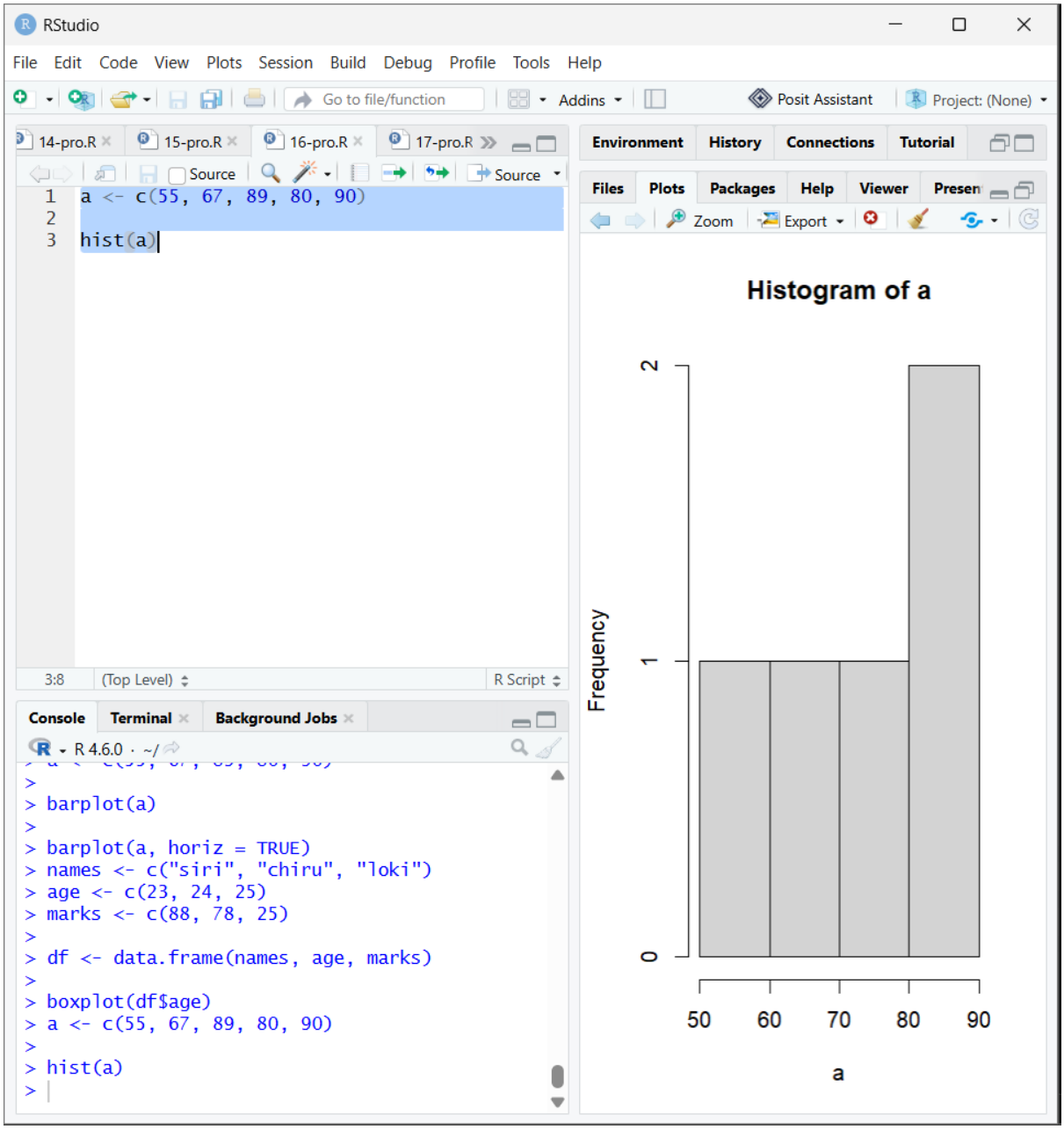
EXP-14


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
12-pro.R
13-pro.R
14-pro.R
15-pro.R
Source
Source
1 a <- c(55, 67, 89, 80, 90)
2
3 barplot(a)
4
5 barplot(a, horiz = TRUE)
5:25
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
[1] 20
> Maximum <- max(A)
> print(Maximum)
[1] 40
> MinMax <- (A - Minimum) / (Maximum - Minimum)
> print(MinMax)
[1] 0.00 0.25 0.50 0.75 1.00
> a <- c(55, 67, 89, 80, 90)
> barplot(a)
> barplot(a, horiz = TRUE)
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Zoom
Export
0
20
40
60
80

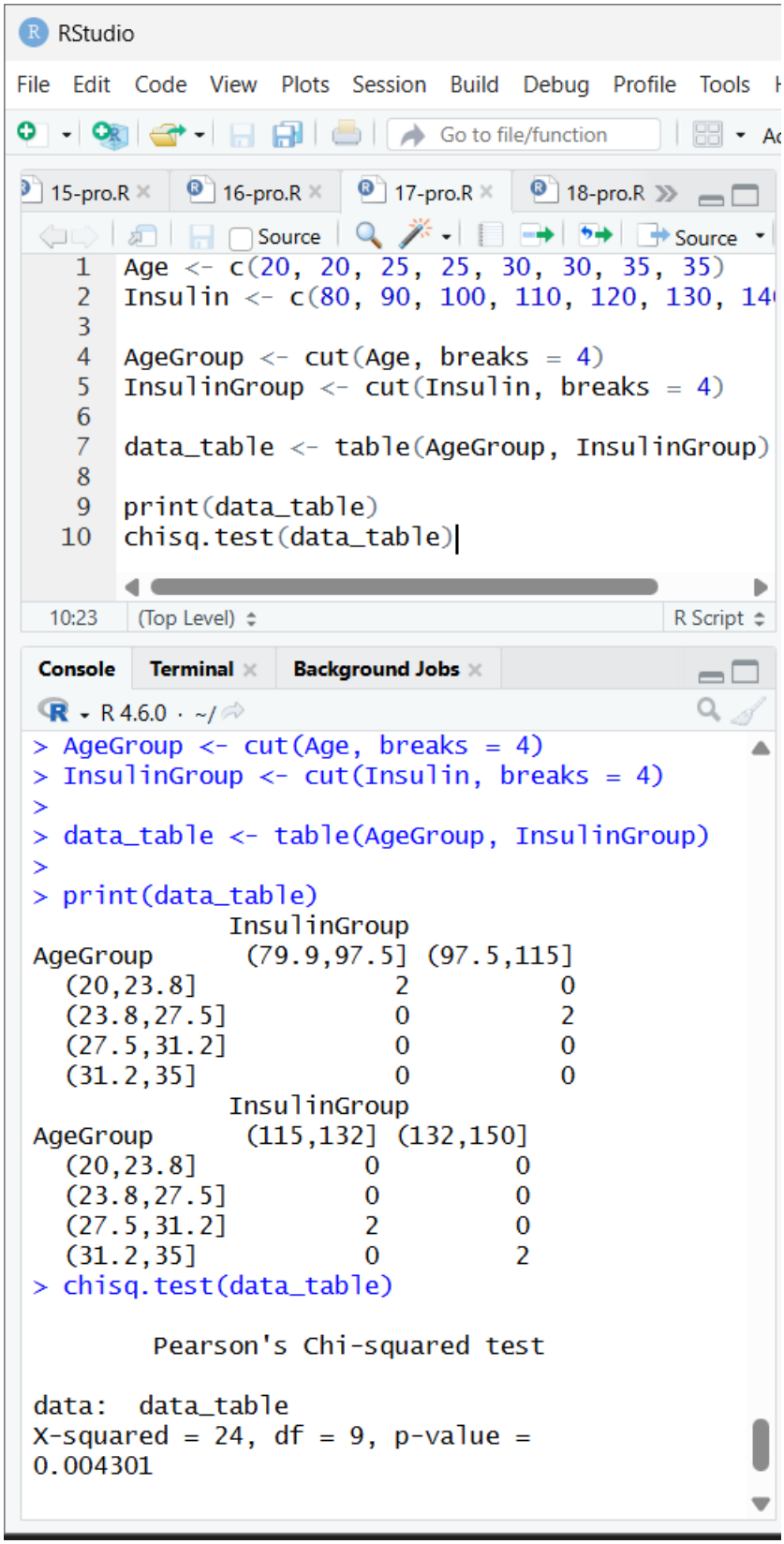
EXP-15


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
13-pro.R
14-pro.R
15-pro.R
16-pro.R
Source
1 names <- c("siri", "chiru", "loki")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 boxplot(df$age)
7:16
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> print(MinMax)
[1] 0.00 0.25 0.50 0.75 1.00
> a <- c(55, 67, 89, 80, 90)
>
> barplot(a)
>
> barplot(a, horiz = TRUE)
> names <- c("siri", "chiru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> boxplot(df$age)
>
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presen
Zoom
Export
25.0
24.5
24.0
23.5
23.0

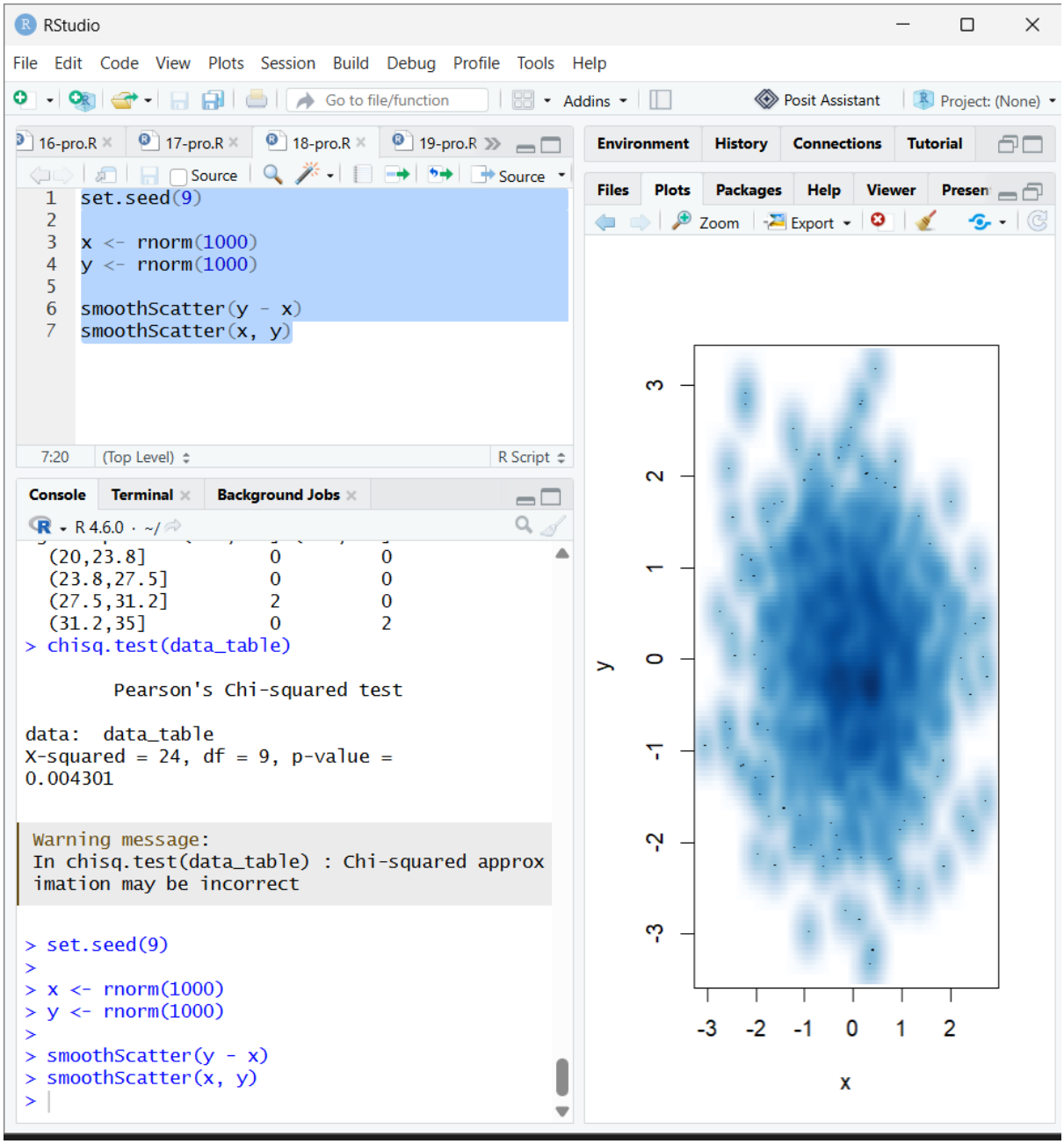
EXP-16


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
14-pro.R
15-pro.R
16-pro.R
17-pro.R
Source
Source
1 a <- c(55, 67, 89, 80, 90)
2
3 hist(a)
3:8
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> barplot(a)
>
> barplot(a, horiz = TRUE)
> names <- c("siri", "chiru", "loki")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> boxplot(df$age)
> a <- c(55, 67, 89, 80, 90)
>
> hist(a)
>
Environment
History
Connections
Tutorial
Files
Plots
Packages
Help
Viewer
Presen
Zoom
Export
Histogram of a
Frequency
0
1
2
50
60
70
80
90
a

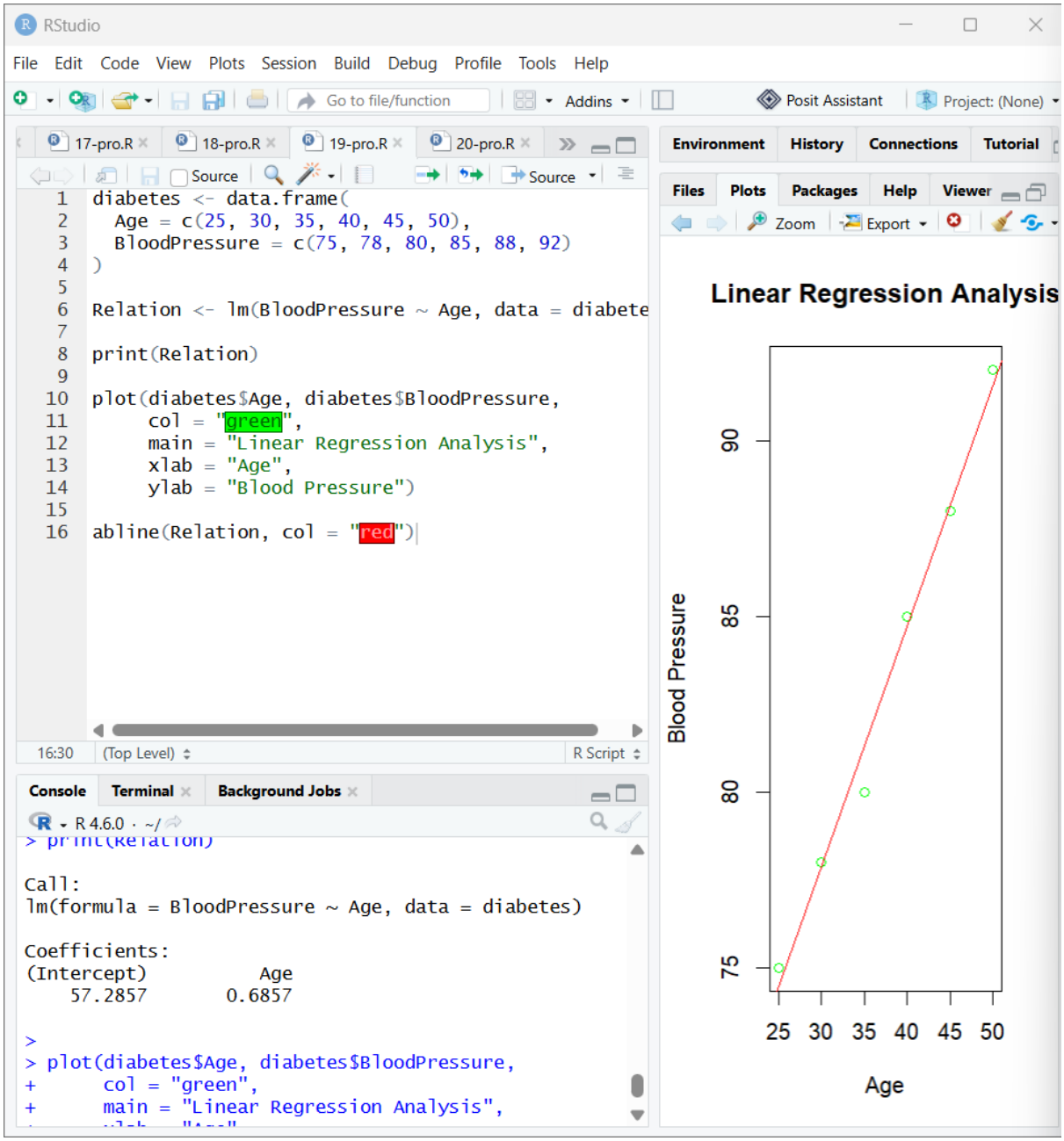
EXP-17


RStudio
File Edit Code View Plots Session Build Debug Profile Tools
Go to file/function
15-pro.R
16-pro.R
17-pro.R
18-pro.R
Source
Source
1 Age <- c(20, 20, 25, 25, 30, 30, 35, 35)
2 Insulin <- c(80, 90, 100, 110, 120, 130, 14
3
4 AgeGroup <- cut(Age, breaks = 4)
5 InsulinGroup <- cut(Insulin, breaks = 4)
6
7 data_table <- table(AgeGroup, InsulinGroup)
8
9 print(data_table)
10 chisq.test(data_table)
10:23
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> AgeGroup <- cut(Age, breaks = 4)
> InsulinGroup <- cut(Insulin, breaks = 4)
>
> data_table <- table(AgeGroup, InsulinGroup)
>
> print(data_table)
InsulinGroup
AgeGroup (79.9,97.5] (97.5,115]
(20,23.8] 2 0
(23.8,27.5] 0 2
(27.5,31.2] 0 0
(31.2,35] 0 0
InsulinGroup
AgeGroup (115,132] (132,150]
(20,23.8] 0 0
(23.8,27.5] 0 0
(27.5,31.2] 2 0
(31.2,35] 0 2
> chisq.test(data_table)
Pearson's Chi-squared test
data: data_table
X-squared = 24, df = 9, p-value =
0.004301

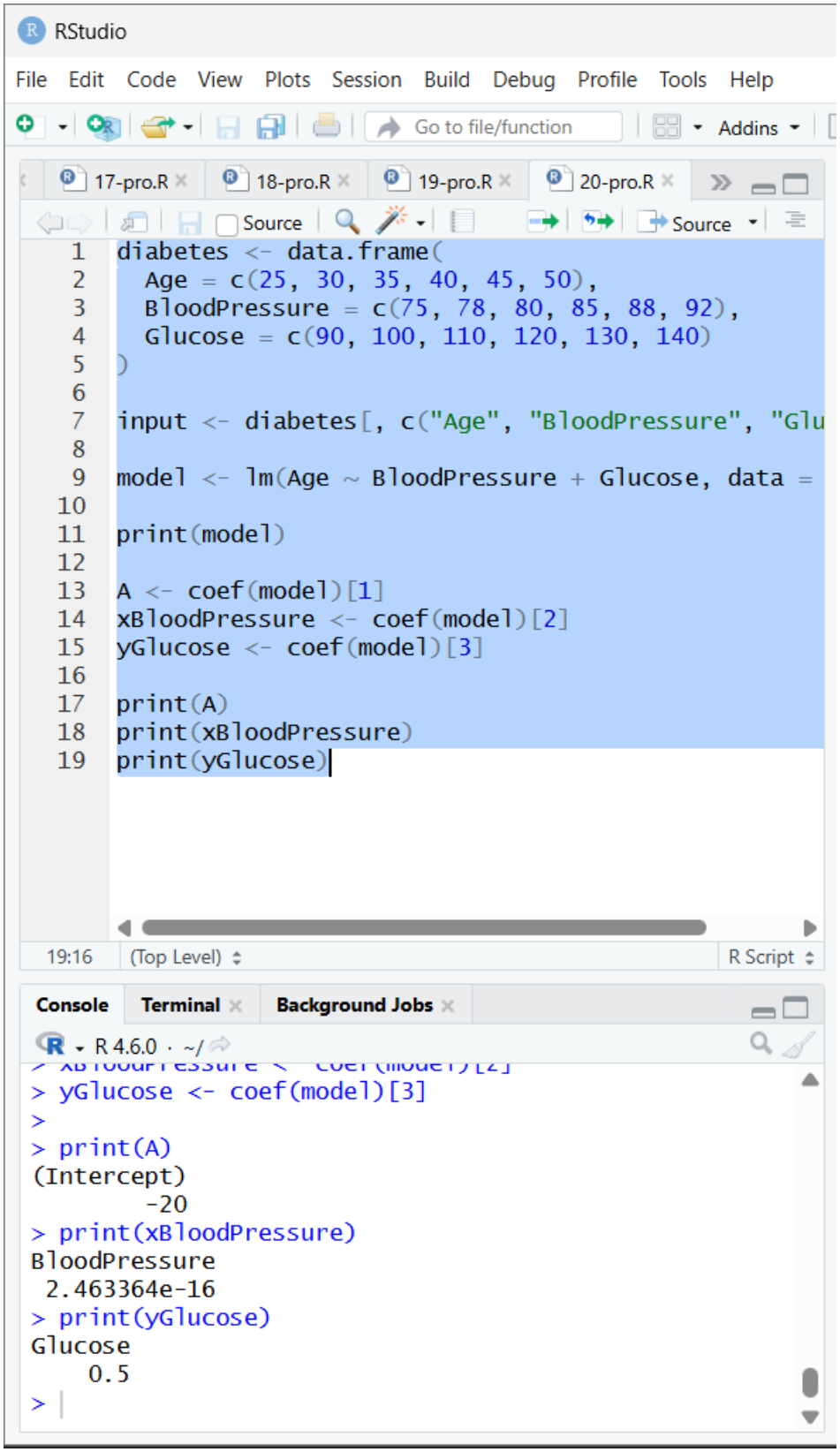
EXP-18


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
16-pro.R
17-pro.R
18-pro.R
19-pro.R
Source
Source
1 set.seed(9)
2
3 x <- rnorm(1000)
4 y <- rnorm(1000)
5
6 smoothScatter(y - x)
7 smoothScatter(x, y)
7:20
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
(20,23.8] 0 0
(23.8,27.5] 0 0
(27.5,31.2] 2 0
(31.2,35] 0 2
> chisq.test(data_table)
Pearson's Chi-squared test
data: data_table
X-squared = 24, df = 9, p-value = 0.004301
Warning message:
In chisq.test(data_table) : Chi-squared approximation may be incorrect
> set.seed(9)
>
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> smoothScatter(y - x)
> smoothScatter(x, y)
>
Environment
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y
x
3
2
1
0
-1
-2
-3
-3 -2 -1 0 1 2

EXP-19


RStudio
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17-pro.R
18-pro.R
19-pro.R
20-pro.R
diabetes <- data.frame(
Age = c(25, 30, 35, 40, 45, 50),
BloodPressure = c(75, 78, 80, 85, 88, 92)
)
Relation <- lm(BloodPressure ~ Age, data = diabete
print(Relation)
plot(diabetes$Age, diabetes$BloodPressure,
col = "green",
main = "Linear Regression Analysis",
xlab = "Age",
ylab = "Blood Pressure")
abline(Relation, col = "red")
Console
Terminal
Background Jobs
Call:
lm(formula = BloodPressure ~ Age, data = diabetes)
Coefficients:
(Intercept) Age
57.2857 0.6857
> plot(diabetes$Age, diabetes$BloodPressure,
+ col = "green",
+ main = "Linear Regression Analysis",
Environment
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Export
Linear Regression Analysis
Blood Pressure
Age
75
80
85
90
25
30
35
40
45
50

EXP-20


RStudio
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Go to file/function
Addins
17-pro.R
18-pro.R
19-pro.R
20-pro.R
Source
Source
diabetes <- data.frame(
Age = c(25, 30, 35, 40, 45, 50),
BloodPressure = c(75, 78, 80, 85, 88, 92),
Glucose = c(90, 100, 110, 120, 130, 140)
)
input <- diabetes[, c("Age", "BloodPressure", "Glu
model <- lm(Age ~ BloodPressure + Glucose, data =
print(model)
A <- coef(model)[1]
xBloodPressure <- coef(model)[2]
yGlucose <- coef(model)[3]
print(A)
print(xBloodPressure)
print(yGlucose)
19:16
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> yGlucose <- coef(model)[3]
>
> print(A)
(Intercept)
-20
> print(xBloodPressure)
BloodPressure
2.463364e-16
> print(yGlucose)
Glucose
0.5
>